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NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2026
Even Semester - 2025-26
PH201 - ENGINEERING PHYSICS

Time: 3 Hrs

Maximum Marks : 60

Instructions to the candidate:*Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. What happens to the resistance of a pure semiconductor when heated?(1) C02 BL1 PO2
(a) The resistance increases (b) The resistance decreases
(c) The temperature remains the same (d) None of these
2. Calculate the relative refractive index difference for a fiber where the core index is 1.50 and the cladding index is 1.45. (1) C01 BL2 PO1
(a) 0.050 (b) 0.012
(c) 0.250 (d) 0.033
3. A key characteristic of laser light is its: (1) C01 BL1 PO1
(a) High directionality (b) Polychromatic nature
(c) Incoherence (d) Low intensity
4. The wavefunction of Fermions is ____; while for Bosons, it is _____. (1) C04 BL1 PO1
(a) Antisymmetric; Symmetric (b) Symmetric; Antisymmetric
(c) Constant; Variable (d) Real; Imaginary
5. The number of lattice points per unit cell in a face-centered cubic (FCC) lattice is: (1) C02 BL1 PO1

- (a) 1 (b) 2
(c) 4 (d) 8
6. The depletion region in a PN junction diode contains: (1) C02 BL1 PO2
(a) A large number of free electrons and holes. (b) Conductors.
(c) Only majority carriers. (d) Primarily immobile ions.
7. In case of p-n junction, if the reverse bias is increased, the width of depletion region (1) C01 BL1 PO1
(a) increases (b) decreases
(c) remains constant (d) none of these
8. The velocity with which energy is transported through a medium is (1) C01 BL2 PO2
(a) Group velocity (b) Phase velocity
(c) Complex velocity (d) Imaginary velocity
9. Size confinement is maximum in (1) C03 BL3 PO3
(a) Bulk material (b) Quantum dot
(c) Quantum wire (d) Quantum well
10. What type of storage is HDD? (1) C01 BL2 PO1
(a) Optical (b) Magnetic
(c) Solid state (d) Mechanical
11. Calculate the de-Broglie wavelength of an electron which has been accelerated from rest on application of potential of 400 volts. (1) C02 BL2 PO2
(a) $0.6135 \text{ \AA}$ (b) $0.1653 \text{ \AA}$
(c) $0.5125 \text{ \AA}$ (d) $0.2514 \text{ \AA}$
12. The miller indices of a plane having intercepts 1,1,4 units along x,y,z axis respectively are (1) C02 BL2 PO2
(a) (4,4,1) (b) (2,3,4)
(c) (4,1,1) (d) (4,4,3)

Part B**(Answer any four of the following) 5x4=20**

13. Describe the construction and working principle of Ruby Laser with energy level diagrams. (5) C01 BL2 PO2
14. (a) Define energy band gap. (1) C02 BL1 PO2

(b) Write down differences between conductors, semiconductors, and insulators based on the energy band diagram.	(4)	C02	BL2	PO2
15. (a) Define nanomaterials.	(1)	C05	BL1	PO1
(b) Discuss their unique properties compared to bulk materials.	(4)	C05	BL2	PO2
16. (a) State Bragg's law of Diffraction.	(2)	C01	BL1	PO1
(b) The spacing between principle planes of NaCl crystal is 2.82 Å. It is found that first order order Bragg reflection occurs at an angle of 100. What is the wavelength of incident ray?	(3)	C01	BL2	PO3
17. X-rays of wavelength 1.54 Å are diffracted from a crystal. The first-order (n=1) reflection occurs at an angle of 15.5°. Calculate the interplanar spacing .	(5)	C02	BL4	PO2
18. (a) Define numerical aperture of a fiber .	(1)	C01	BL1	PO1
(b) Find out expression for the same in step –index optical fiber ?	(2)	C02	BL2	PO2
(c) Find the numerical aperture and angle of acceptance of a given optical fiber if the refractive indices of the core and cladding are 1.54 and 1.50 respectively.	(2)	C02	BL3	PO2

Part C**(Answer any three of the following) 3x10=30**

19. (a) Differentiate between p-type and n-type semiconductors concerning their doping elements	(5)	C02	BL2	PO2
(b) Calculate Atomic Packing Factor (APF) for a simple cubic structure.	(5)	C02	BL2	PO2
20. (a) Describe Quantum confinement.	(2)	C05	BL2	PO3
(b) Distinguish between Quantum well, Quantum wire and Quantum dots.	(5)	C05	BL2	PO3
(c) Explain the construction and application of LED.	(3)	C05	BL1	PO1
21. (a) Derive Einstein A, B co efficient relation. State the physical significance of the of Einstein A,B coefficient relation.	(8)	C01	BL3	PO2
(b) Explain the working principle of a semiconductor laser.	(2)	C01	BL2	PO2

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| 22. (a) State Heisenberg's uncertainty principle. | (2) | C03 | BL1 | PO1 |
| (b) Distinguish between Boson and Fermion. | (3) | C04 | BL2 | PO2 |
| (c) Calculate the de Broglie wave-length of an electron moving through potential difference of 15 Volt. | (5) | C03 | BL4 | PO3 |
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| 23. (a) Compare M-B, F-D and B-E statistics mentioning at least three characteristics . | (5) | C02 | BL3 | PO1 |
| (b) Draw the fermi distribution curve of (i) $T=0K$ (ii) $T \neq 0K$. Explain their significance. | (5) | C03 | BL4 | PO2 |